EC417 Mini Project P07

Wine Quality Classification — Final Results

Problem and metric

Predict quality ≥ 7 from physicochemical measures in red and white Vinho Verde wines; this tests how lab measures proxy subjective sensory judgments. Only 19.0% of cleaned rows are positive, so PR-AUC (average precision) is primary: it foregrounds positive-class retrieval and reflects the low base rate. Balanced accuracy averages sensitivity and specificity at a particular threshold. The BA @ 0.5 column applies the same fixed cutoff to every model; its mean values are LR 0.759, RF 0.638, and GB 0.761. The validation-threshold BA uses each model's validation-selected cutoff; these are distinct operating points, not ranking scores.

Data and protocol

UCI Wine Quality; CC BY 4.0. Rows before/after exact de-duplication: 6,497 → 5,320. Rows involved in duplicate groups (keep=False): 2,169; redundant rows removed (keep='first'): 1,177. Each of 5 seeds uses a stratified 60%/20%/20% train/validation/test split after de-duplication. Indices always resolve against the cleaned dataframe.

Method

Dummy prior baseline; logistic regression reference; random forest main model and histogram gradient boosting comparator. Configurations use validation PR-AUC; the sole operating-point improvement is to select threshold by validation balanced accuracy. The 11 physicochemical columns are inputs; is_red is not.

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Raw red + white rows → de-duplicate → stratified split → select model on validation
                                                       → choose threshold on validation → test once
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Results — mean ± SD, 5 seeds

Model	BA @ 0.5	BA @ val.	ROC-AUC	PR-AUC	Fit s
Dummy prior	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000	0.190 ± 0.000	0.001 ± 0.000
Logistic regression	0.759 ± 0.013	0.756 ± 0.010	0.817 ± 0.008	0.499 ± 0.017	0.015 ± 0.001
Random forest	0.638 ± 0.014	0.783 ± 0.011	0.847 ± 0.009	0.565 ± 0.018	2.889 ± 0.154
Gradient boosting	0.761 ± 0.011	0.764 ± 0.009	0.836 ± 0.006	0.531 ± 0.024	0.460 ± 0.151

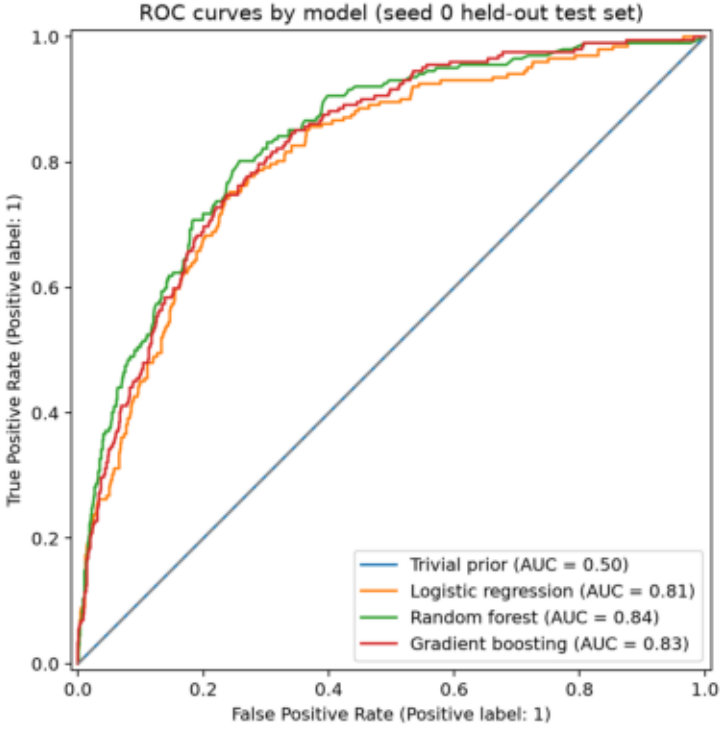
Validation-threshold BA means are LR 0.756, RF 0.783, and GB 0.764. The seed-0 RF validation threshold is 0.19. The RF uses balanced_subsample class weights; its probabilities were not calibrated to a 0.5 cutoff here, and calibration was not tested. A reliability diagram could assess calibration; no cause for the threshold difference is claimed. RF exceeded GB in PR-AUC on every paired seed (RF−GB: +0.021, +0.037, +0.050, +0.034, +0.031). GB early stopping uses sklearn's internal validation split; iterations by seed: 235, 44, 38, 138, 90.

See README.md for complete per-seed tables and secondary analyses.

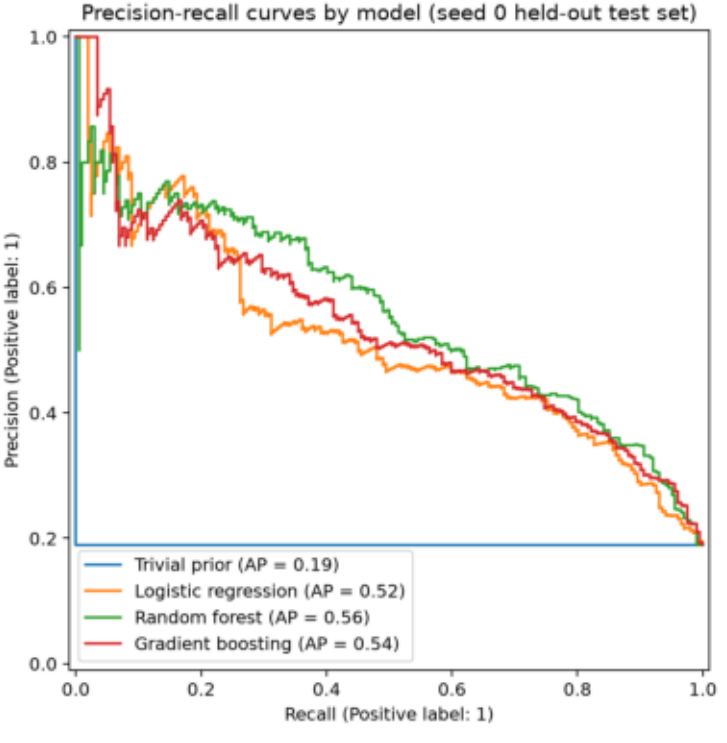
Results: held-out ranking curves

All models are overlaid on the seed-0 test set; selection and threshold choice used validation only. RF−GB paired PR-AUC differences: +0.021, +0.037, +0.050, +0.034, +0.031. Possible explanations—not facts—include the small de-duplicated sample, noisy sensory labels, unstable GB early stopping, and the small tuning budget.

ROC curves



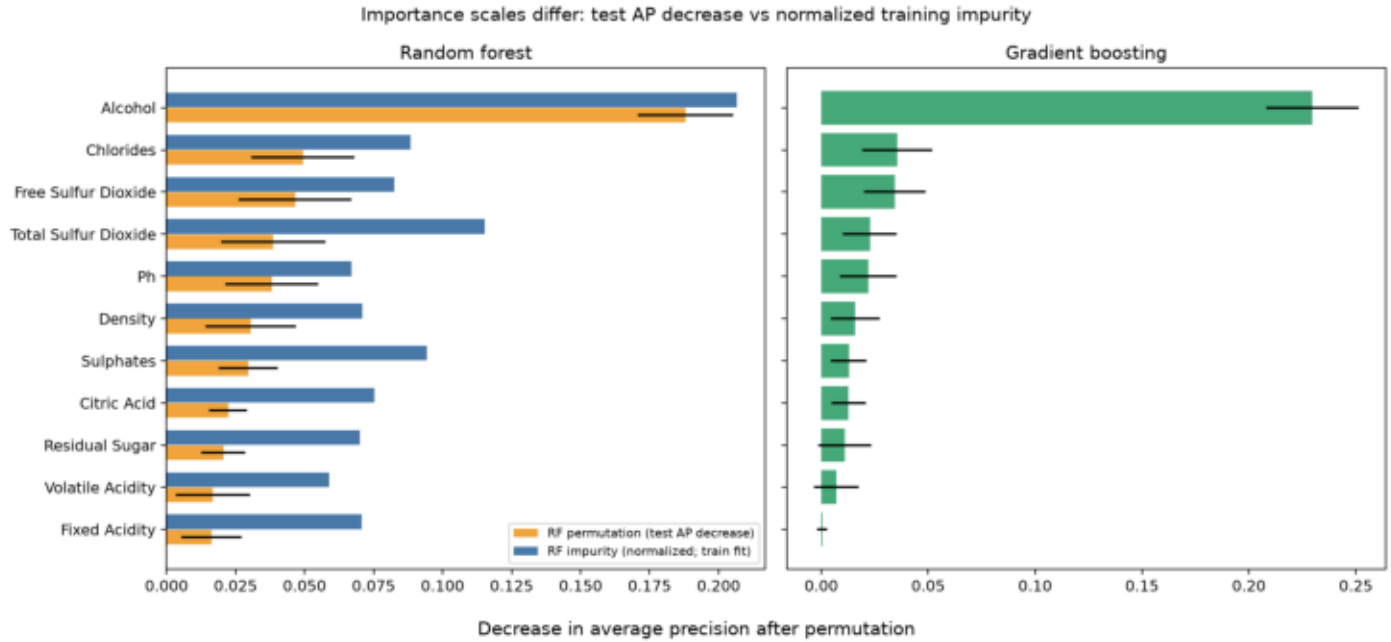
Precision-recall curves



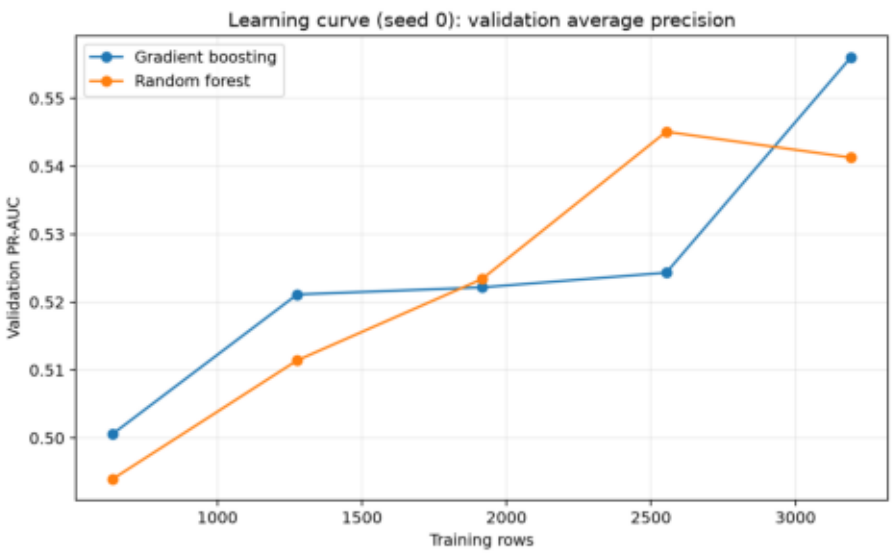
Feature importance and model diagnostics

Permutation importance is the test-set average-precision decrease (10 repeats); RF impurity importance is a normalized training-fit scale. Compare rankings within a method, not magnitudes across methods.

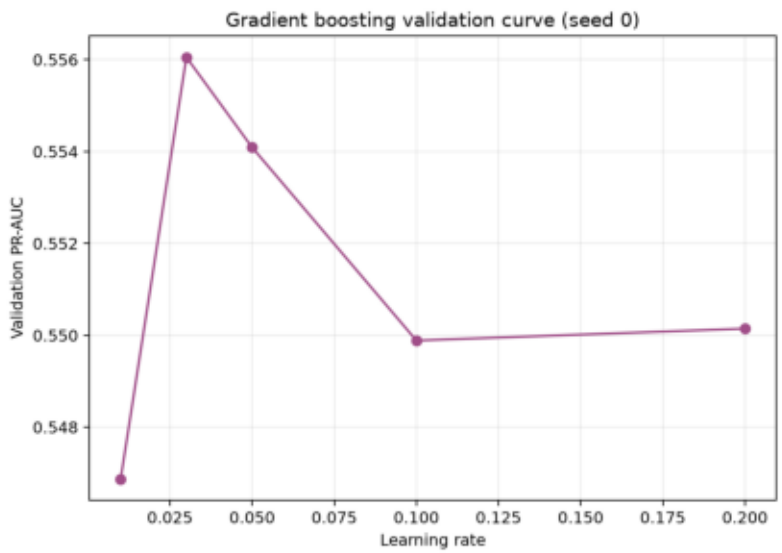
Permutation and impurity importance



Learning curve



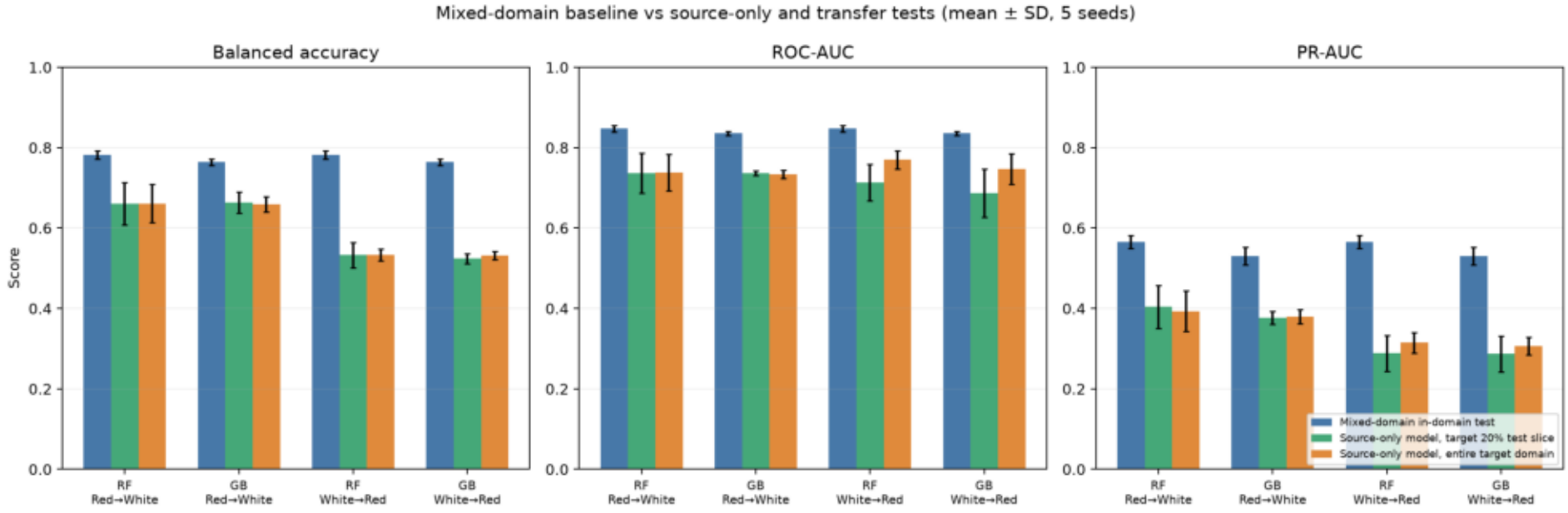
GB learning-rate validation curve



Domain shift: source-only models and target domains

Threshold is selected on source validation and fixed for target evaluation. The figure contrasts the mixed-domain test reference, the original target 20% test slice, and the entire cleaned target domain. Compare PR-AUC with its lift over target prevalence because base rates differ.

Mean ± SD over the saved seeds



Source-only model	BA	ROC-AUC	PR-AUC	PR-AUC lift	Target rate	Positives / rows
Random forest, red → white	0.661 ± 0.048	0.738 ± 0.046	0.394 ± 0.051	1.889 ± 0.245	0.208	825/3961
Gradient boosting, red → white	0.658 ± 0.019	0.734 ± 0.010	0.380 ± 0.018	1.823 ± 0.085	0.208	825/3961
Random forest, white → red	0.533 ± 0.015	0.770 ± 0.022	0.315 ± 0.026	2.324 ± 0.188	0.135	184/1359
Gradient boosting, white → red	0.532 ± 0.010	0.746 ± 0.038	0.307 ± 0.022	2.265 ± 0.161	0.135	184/1359

Red→white ROC-AUC falls by 0.109 (RF) and 0.101 (GB) versus mixed-domain results; RF/GB target ranges overlap, so no transfer winner is claimed. White→red BA is 0.533 (RF) / 0.532 (GB), near chance, while ROC-AUC is 0.770 / 0.746. Largest distribution shifts: total sulfur dioxide, volatile acidity, free sulfur dioxide. These feature shifts are hypotheses, not causal findings.

Error analysis — seed-0 random forest

Validation threshold 0.190; 268/1064 errors. Quality 6 or 7: 222 errors (82.8%). Red 58/272 (21.3%); white 210/792 (26.5%). Good-class TP/FP/FN/TN=162/228/40/634; precision=0.415 (share of predicted-good wines that are good), recall=0.802 (share of good wines found).

The ten rows below are five false positives and five false negatives, ranked by distance from the threshold within each group. FP mean alcohol=12.63; FN mean alcohol=9.46. FP alcohol is higher on average, consistent with alcohol being a dominant RF signal (permutation-importance rank 1); this is based on only ten examples and does not establish cause.

Err	Type	Q	P(good)	Dist.	Fixed acid.	Volatile acid.	Citric acid	Residual sugar	Chlor.	Free SO2	Total SO2	Density	pH	Sulph.	Alcohol
FP	White	6	0.877	0.688	6.6	0.25	0.35	2.9	0.034	38	121	0.99	3.19	0.4	12.8
FP	White	6	0.845	0.655	5.7	0.28	0.3	3.9	0.026	36	105	0.99	3.26	0.58	12.8
FP	White	6	0.826	0.636	5.8	0.42	0.3	2.2	0.035	26	129	0.989	3.32	0.47	12.9
FP	White	6	0.814	0.624	6.4	0.31	0.28	2.5	0.039	34	137	0.989	3.22	0.38	12.7
FP	Red	6	0.797	0.607	10.5	0.36	0.47	2.2	0.074	9	23	0.996	3.23	0.76	12
FN	Red	7	0	0.19	7.5	0.52	0.16	1.9	0.085	12	35	0.997	3.38	0.62	9.5
FN	White	7	0	0.19	6	0.34	0.24	5.4	0.06	23	126	0.995	3.25	0.44	9
FN	Red	7	0.0112	0.179	7.8	0.58	0.02	2	0.073	9	18	0.997	3.36	0.57	9.5
FN	Red	7	0.0125	0.177	7.3	0.48	0.32	2.1	0.062	31	54	0.997	3.3	0.65	10
FN	White	7	0.015	0.175	7	0.3	0.38	14.9	0.032	60	181	0.998	3.18	0.61	9.3

Labels and feature values join through the cleaned sample_index; code asserts actual_good == (quality >= 7) for each row.

Limitations, reproducibility, and next steps

Limitations and ethics

- Exact duplicates were removed; near-duplicates and source-level dependence may remain.
- Sensory labels are subjective/noisy; a binary threshold discards ordinal information.
- This is one Vinho Verde dataset. The original red target slice has only 37 positives.
- Seeds overlap in their samples, so seed SD understates sampling uncertainty.
- Tuning grids are small; GB early stopping varies substantially and uses sklearn's internal split.
- is_red is not a feature. Educational use only; no commercial or human-impact decision use.

Reproduce

From a fresh clone: python3 -m venv .venv; .venv/bin/pip install -r requirements.txt; ./run_all.sh. This downloads UCI data, trains the saved seeds, regenerates CSV/JSON results and figures, writes README.md, and builds this report.

Fresh-clone total (clone, environment, install, run): 383 seconds (6.4 minutes). Hardware/software: 32 logical CPUs, 30.8 GiB RAM; Linux-4.18.0-514.el8.x86_64-x86_64-with-glibc2.28; Python 3.14.7. Python 3.11.4 requirements install failed: Pinned contourpy==1.4.0 is incompatible: pip reports Requires-Python >=3.12.

What I would try next

- Allocate a larger, predeclared validation budget to GB hyperparameter tuning.
- Use a cross-validated early-stopping scheme that respects the project's split.
- Evaluate probability calibration before choosing a probability-based operating threshold.
- Treat quality as ordinal and compare ordinal regression against this binary target.

AI use — author TODO

The coding assistant was an AI assistant using Copilot SDK in VS Code. It helped create the code scaffold, run experiments, generate the README and report, and fix errors. Claude (Anthropic) reviewed results, spotted issues (including the index-alignment bug and weighted-RF wording), and provided advice.

TODO (student): list what I personally verified, for example the leakage-free split, the label/index alignment fix, the numbers in the results table, and the viva answers.

References

Cortez et al. (2009), Decision Support Systems 47(4), 547–553, doi:10.1016/j.dss.2009.05.016; Breiman (2001), Machine Learning 45, 5–32; Friedman (2001), Annals of Statistics 29(5), 1189–1232; scikit-learn documentation for RandomForestClassifier, HistGradientBoostingClassifier, permutation importance, and average precision.